

70 Important Python, Data Analysis, and Data Science Interview Questions

Layman definitions, analogies, and examples for fast interview preparation.

Interview Formula

For each answer, say: definition first, then analogy, then one practical example. This makes your answer clear and easy to remember.

Python Basics

1. What is data science?

Definition: Data science means using data to understand problems, find patterns, make predictions, and support decisions.

Analogy: Like detective work: collect clues, clean them, connect them, and explain the story.

Example: A bank predicts which customers may not repay a loan.

2. What is the difference between data analysis and data science?

Definition: Data analysis explains what happened. Data science often also predicts what may happen next.

Analogy: Analysis is reading yesterday's match score. Data science is predicting tomorrow's winner.

Example: Sales analysis shows last month's revenue; data science predicts next month's revenue.

3. What is Python used for in data science?

Definition: Python is used for cleaning data, analysis, visualization, machine learning, automation, and reporting.

Analogy: It is a toolbox with many tools inside.

Example: pandas cleans tables, matplotlib draws charts, scikit-learn trains models.

4. What is a variable in Python?

Definition: A variable is a name that stores or points to a value.

Analogy: Like a label on a box.

Example: age = 25 stores 25 under the name age.

5. What is a data type?

Definition: A data type tells Python what kind of value something is.

Analogy: Like containers: bottle for water, box for books.

Example: int, float, string, list, tuple, dict, set, bool.

6. Mutable vs immutable?

Definition: Mutable values can change after creation. Immutable values cannot change in place.

Analogy: Mutable is a whiteboard; immutable is printed paper.

Example: list is mutable; tuple and string are immutable.

7. List vs tuple?

Definition: A list can change; a tuple is fixed.

Analogy: List is a shopping cart; tuple is a sealed package.

Example: Use list for changing features; tuple for fixed coordinates.

8. Set vs dictionary?

Definition: A set stores unique values. A dictionary stores key-value pairs.

Analogy: Set is a guest list; dictionary is a phonebook.

Example: {'A', 'B'} vs {'A': 90, 'B': 85}.

9. What is a function?

Definition: A function is reusable code that performs a task.

Analogy: Like a machine: input goes in, output comes out.

Example: def add(a, b): return a + b.

10. What is a lambda function?

Definition: A lambda is a small one-line function without a normal name.

Analogy: Like a sticky note calculation.

Example: `lambda x, y: x * y` multiplies two values.

11. Explain lambda x, y: x*y.

Definition: x and y are inputs; x*y is returned. If x=7 and y=19, the answer is 133.

Analogy: Like saying 'take two numbers and multiply them' in one sentence.

Example: `a = lambda x,y: x*y`; `a(7,19)` gives 133.

12. What are *args?

Definition: *args collects extra positional arguments into a tuple.

Analogy: Like a bag of unnamed items.

Example: `def total(*args): return sum(args)`.

13. What are **kwargs?

Definition: **kwargs collects extra named arguments into a dictionary.

Analogy: Like labeled boxes.

Example: `def show(**kwargs): print(kwargs)`.

14. What is a local variable?

Definition: A local variable is created inside a function and used inside that function.

Analogy: Like money in your pocket.

Example: message inside `def greet()` is local.

15. What is a global variable?

Definition: A global variable is created outside functions and can be accessed more widely.

Analogy: Like money on a shared table.

Example: `name = 'Rita'` outside a function is global.

16. What is range()?

Definition: `range()` generates numbers for looping.

Analogy: Like a ticket machine giving numbers one by one.

Example: `for i in range(5)` gives 0 to 4.

17. What is xrange()?

Definition: `xrange` existed in Python 2. In Python 3 it is removed; `range` is already memory efficient.

Analogy: Old `xrange` was like a lazy ticket machine. Python 3 `range` does that job.

Example: Use `range()` in Python 3.

18. What is list comprehension?

Definition: A short way to create a list from another iterable.

Analogy: Like an assembly line.

Example: `[x*x for x in range(5)]`.

19. What is dictionary comprehension?

Definition: A short way to create a dictionary.

Analogy: Like making labels for boxes quickly.

Example: {x: x*x for x in range(3)}.

20. What is exception handling?

Definition: It handles errors using try, except, and finally.

Analogy: Like driving with insurance and cleanup.

Example: try reading file, except file missing, finally close file.

Data Analysis With Python

21. What is pandas?

Definition: pandas is a Python library for table data.

Analogy: Like Excel controlled by code.

Example: `df = pd.read_csv('data.csv')`.

22. What is a DataFrame?

Definition: A DataFrame is a table with rows and columns.

Analogy: Like a spreadsheet.

Example: Customer rows and age/salary columns.

23. What is a Series?

Definition: A Series is one labeled column from a DataFrame.

Analogy: Like one column of a spreadsheet.

Example: `df['age']` is a Series.

24. How do you read a CSV file?

Definition: Use pandas `read_csv`.

Analogy: Like opening an Excel file in Python.

Example: `df = pd.read_csv('sales.csv')`.

25. How do you inspect a dataset?

Definition: Use `head`, `info`, `describe`, `shape`, `columns`, and `value_counts`.

Analogy: Like checking a new house room by room.

Example: `df.head()`, `df.info()`, `df.describe()`.

26. What is `df.head()`?

Definition: It shows the first few rows of a DataFrame.

Analogy: Like previewing the first page of a book.

Example: `df.head(10)` shows first 10 rows.

27. What is `df.info()`?

Definition: It shows column names, missing values, and data types.

Analogy: Like a medical report for your dataset.

Example: Use it before cleaning data.

28. What is `df.describe()`?

Definition: It gives summary statistics for numeric columns.

Analogy: Like a quick scorecard.

Example: Mean, min, max, quartiles.

29. How do you handle missing values?

Definition: Drop them, fill them, predict them, or mark them with a flag.

Analogy: Like blank form answers: remove, estimate, or label as blank.

Example: `df['age'].fillna(df['age'].median())`.

30. Mean vs median?

Definition: Mean is average. Median is the middle value after sorting.

Analogy: Mean can be pulled by one rich person; median is more stable.

Example: For salaries, median is often better.

31. How do you handle outliers?

Definition: Check if they are errors or real rare values, then keep, cap, transform, or remove.

Analogy: Like one loud voice in a quiet room.

Example: Cap extreme income values at the 99th percentile.

32. What is duplicate data?

Definition: Duplicate data means repeated rows or records.

Analogy: Like the same person signing the attendance sheet twice.

Example: `df.drop_duplicates()`.

33. What is groupby?

Definition: groupby splits data into groups and calculates summaries.

Analogy: Like sorting students by class and finding average marks.

Example: `df.groupby('city')['sales'].sum()`.

34. What is a join/merge?

Definition: It combines tables using a common column.

Analogy: Like matching student names with their marks table.

Example: `pd.merge(customers, orders, on='customer_id')`.

35. Inner join vs left join?

Definition: Inner keeps matching rows only. Left keeps all rows from the left table.

Analogy: Inner is guests with tickets only; left is all invited guests, with ticket info if found.

Example: Use left join when the main table must stay complete.

36. What is data visualization?

Definition: It means using charts to understand and communicate data.

Analogy: Like turning numbers into a picture.

Example: Bar chart for sales by region.

37. Which chart for comparison?

Definition: Use bar charts for comparing categories.

Analogy: Like comparing heights of different buildings.

Example: Sales by product category.

38. Which chart for trend?

Definition: Use line charts for changes over time.

Analogy: Like seeing a road going up or down.

Example: Monthly revenue trend.

39. Which chart for distribution?

Definition: Use histogram or boxplot.

Analogy: Like seeing how exam scores are spread.

Example: Distribution of customer age.

40. What is correlation?

Definition: Correlation measures how two variables move together.

Analogy: Umbrella sales and rain may rise together.

Example: Correlation does not prove causation.

41. What is feature engineering?

Definition: Creating useful input columns from raw data.

Analogy: Like preparing ingredients before cooking.

Example: Create day_of_week from date.

42. What is one-hot encoding?

Definition: Converting categories into 0/1 columns.

Analogy: Like checkboxes for each category.

Example: Color red/blue becomes color_red and color_blue.

43. What is label encoding?

Definition: Converting categories into numbers.

Analogy: Like assigning roll numbers to names.

Example: small=0, medium=1, large=2.

44. What is feature scaling?

Definition: Putting numeric features on similar ranges.

Analogy: Like converting meters and kilometers before comparing.

Example: StandardScaler for logistic regression or KNN.

45. Normalization vs standardization?

Definition: Normalization often scales 0 to 1. Standardization makes mean 0 and standard deviation 1.

Analogy: Normalization fits into a fixed box; standardization measures distance from average.

Example: MinMaxScaler vs StandardScaler.

Machine Learning And Data Science

46. What is supervised learning?

Definition: Learning from data that has target answers.

Analogy: Like practicing with an answer key.

Example: Predict house price from past house sales.

47. What is unsupervised learning?

Definition: Finding patterns without target answers.

Analogy: Like sorting mixed clothes without labels.

Example: Customer clustering.

48. Classification vs regression?

Definition: Classification predicts categories. Regression predicts numbers.

Analogy: Label vs price tag.

Example: Spam/not spam vs house price.

49. What is train-test split?

Definition: Training on one part and testing on unseen data.

Analogy: Like studying practice questions and taking a separate exam.

Example: 80% train, 20% test.

50. What is cross-validation?

Definition: Testing the model on several different splits.

Analogy: Like taking multiple mock tests.

Example: 5-fold cross-validation.

51. What is overfitting?

Definition: Model memorizes training data and fails on new data.

Analogy: Like memorizing homework answers.

Example: Training accuracy high, test accuracy low.

52. What is underfitting?

Definition: Model is too simple and misses the pattern.

Analogy: Like explaining a movie with one word.

Example: Linear model for complex curved data.

53. Bias vs variance?

Definition: Bias is error from simplicity; variance is error from sensitivity.

Analogy: Bias is stubborn; variance is nervous.

Example: Linear model high bias, deep tree high variance.

54. What is data leakage?

Definition: Using information during training that will not exist at prediction time.

Analogy: Like giving the answer key before exam.

Example: Using future cancellation date to predict churn.

55. What is accuracy?

Definition: Correct predictions divided by total predictions.

Analogy: Like percentage of exam answers correct.

Example: Good only when classes are balanced.

56. What is precision?

Definition: Of predicted positives, how many are truly positive.

Analogy: Careful fishing: most caught fish are correct.

Example: Useful when false positives are costly.

57. What is recall?

Definition: Of actual positives, how many were found.

Analogy: Wide net: catch as many positives as possible.

Example: Important in disease detection.

58. What is F1-score?

Definition: A balance between precision and recall.

Analogy: Combined grade for carefulness and coverage.

Example: Useful for imbalanced classification.

59. What is confusion matrix?

Definition: A table of true positives, true negatives, false positives, and false negatives.

Analogy: Like checking correct and wrong security decisions.

Example: Helps explain precision and recall.

60. What is ROC-AUC?

Definition: It measures how well a classifier separates classes across thresholds.

Analogy: Like judging how well a metal detector separates coins from stones.

Example: Higher AUC means better separation.

61. What is MAE?

Definition: Mean Absolute Error is the average absolute prediction error.

Analogy: Average distance from the target.

Example: If house price error averages \$10k, MAE is \$10k.

62. What is RMSE?

Definition: Root Mean Squared Error punishes large errors more strongly.

Analogy: Like giving bigger penalties for bigger misses.

Example: Useful when big mistakes are very bad.

63. What is R-squared?

Definition: It shows how much target variation the model explains.

Analogy: Like how much of the story your model can explain.

Example: $R^2 = 0.80$ means model explains about 80% variation.

64. What is linear regression?

Definition: A model that predicts a number using a straight-line relationship.

Analogy: Drawing the best straight road through points.

Example: Predict salary from experience.

65. What is logistic regression?

Definition: A classification model that predicts probability.

Analogy: A risk meter from 0 to 1.

Example: Churn probability = 0.75.

66. What is decision tree?

Definition: A model that asks a sequence of questions to predict.

Analogy: Like 20 Questions.

Example: If age > 30 and income high, predict approved.

67. What is random forest?

Definition: Many decision trees combined together.

Analogy: Asking many judges and taking a vote.

Example: Often better than one tree.

68. What is gradient boosting?

Definition: Models are built one by one, each fixing previous mistakes.

Analogy: A coach correcting errors after each practice.

Example: XGBoost, LightGBM, CatBoost.

69. What is clustering?

Definition: Grouping similar records without labels.

Analogy: Sorting laundry into similar piles.

Example: Group customers by buying behavior.

70. What is K-means?

Definition: A clustering method that groups data into K clusters.

Analogy: Choosing K meeting points and assigning people to the nearest one.

Example: Segment customers into 3 groups.